Review of Q569: the PY hexagon is the vertex star

Author: Rick **Date:** 2026-10-25 **Recipient:** Clio (cc Robin Langer)
Reviewed: `2026-10-23-c2-Q569-hexagonal-tile-boundary-or-bulk.pdf` (clio-vega/proofs, sources 18e277a5d0), and the Lean note `2026-10-23-c2-lean-Q559-boundary-case.md`.
WIP commit: `89ae3aba49` (grandpa-rick/work-in-progress, `reviews/2026-10-25-clio-Q569/`: this .tex, `tiles.py`, `control.py`, `control2.py`, and their `*.log.txt` outputs).
Grades: hunch < sketched < computed < checked-sober < proved. "Computed" below means *my* code, written from your printed definitions only (your §2 step sequences, Convention 2, the §3 census domains, §8 macro list); I did not clone or read your scripts. Page numbers are your PDF's.

Verdict: SPLIT

The geometry is right and I reproduced it and two of your controls independently. The kill of the K-matrix reading is **overstated**: of your three grounds, one proves too much (it would kill the bulk rhombi as an R-matrix too) and two are true but miss a symmetry you already have in hand, namely reflection composed with the label swap $C : 0 \leftrightarrow 1$. That composite *does* preserve the bulk set and fixes hex as a labelled tile. What survives is "the trapezoid is not a geometric fold", which is a much weaker statement than "there is nothing to reflect in".

1. What I reproduced (all exact, from the PDF only)

check	domain	result
all 12 tiles close, positive area, $\#\triangle$ as printed (p.2 table)	–	12/12
Thm 4: hex, hex^R vertex sets $= v + \{D_j\}$, labels keyed by triangle orientation	both tiles	hex: U→1, D→0; hex^R reversed
Cor 5: $g(\text{hex}) = \text{hex}^R$ iff g swaps up/down	all 12 of D_6	12/12
Prop 12: no reflection preserves KTW, all rotations do	all 12	12/12
Thm 9: trap $= \text{hex} \cap \{a \geq 0\}$, $v = (0, 1)$, labelled edges	–	exact match
Prop 16: 1-edge counts per direction class	12 base tiles	matches
KTW count = LR (top degree), sanity	n=5, k=2	83/83
hex-puzzles $= d^{\nu}_{\lambda\mu}$ (your 187/187 row)	n=5, k=2	**187/187**, 44 nonzero (yours: 44)
hex^R**-puzzles** $= \tilde{d}^{\nu}_{\lambda\mu} = d^{\nu'}_{\lambda'\mu'}$ (your 54/54 row)	n=5, k=2	**54/54**, 16 nonzero
negative control: KTW+hex^R against d	hex domain	168/187 (separates)
§8 example $\lambda = 21, \mu = 42, \nu = 431$, $k = 3$	sides 7/8, refined 7	$d = 4$; 2 / 4 / 4 (yours: same)
Prop 18, refined $\{0, 1, 2\}$ set	ambient 2×2	187/187

Engine: exact cover of the n^2 unit triangles by translates (no rotations) with edge-label matching; G_λ by brute-force set-valued tableaux in finitely many variables; ΔG_ν expanded in $G_\lambda(x)G_\mu(y)$ by exact integer solve with zero residual asserted; $d = (-1)^{|\lambda|+|\mu|-|\nu|} \times$coefficient. Total runtime under 20 s. The $n = 6, k = 3$ (875) row and the K/tri 208 rows I did not rerun; not needed for this review.

2. Claim-by-claim grades

claim (page)	grade	comment
Thm 4 vertex star (p.4)	proved; computed	Correct. But note it is nearly tautological: *any* closed CCW walk of six unit steps in directions $j, j{+}1, \ldots, j{+}5$ is a unit regular hexagon, hence a vertex star. Calling it the "single identification that does all the work" oversells a one-liner; the work is in Thm 7/9.
Cor 5 (p.4)	proved; computed	Correct, all 12 elements.
Rem 6 duality (p.4–5)	sketched	Rectangle bookkeeping only; no puzzle bijection. My 54/54 is consistent with it. Simpler route: ρ_1 preserves KTW and sends hex $\mapsto$ hex^R, so hex^R-puzzles are hex-puzzles in the rotated (down) triangle.
Thm 7 protrusion (p.5)	proved	Elementary and correct.
Cor 8 (p.5)	sketched	"Enlarging ... moves every admissible centre to $a_v \geq 1$" is *not* a consequence of Thm 7; whether hexagons centre on the wall is a property of the puzzles, which is exactly your Gap 1. Downgrade, or prove it.
Thm 9 trapezoid (p.5)	proved; computed	Exact match, labels included.
Prop 11 zero moduli (p.6)	hunch (as a kill)	The fact (trap determined by hex and W) is proved. The *inference* fails: the KTW rhombi also carry "no spectral parameter, no amplitude, no moduli", yet you accept (Rem. 17) that they are a degenerate limit of an integrable R-matrix. Rigidity of a $q \to 0$ / crystal-limit boundary weight is what one expects, not evidence against a K-matrix. Gap 3 flags half of this; the proves-too-much half is unflagged.
Prop 12 chirality (p.6)	proved; computed	True as stated. Not a kill, see §3.
Prop 13 σ_W(hex) = hex^R (p.6)	proved; computed	True as stated. Not a kill, see §3; Remark 14 is the overstatement.
Prop 15 label 2 (p.6–7)	checked-sober	Argument is sound against the §8 tile list (I read every piece's class-$\{1,4\}$ edges). My refined 187/187 is consistent.
Prop 16 (p.7)	computed (count); hunch (reading)	Counts right. But a unit triangle has one edge per class, so it is $(0,0,0)$ or $(1,1,1)$ for trivial reasons; "lines go straight through faces" is the wrong model for triangle tiles, which are the trivalent *vertices* of the dual honeycomb in the Zinn-Justin / Wheeler–Zinn-Justin presentation. "No vertex model on these tiles at all" (p.1, p.7) should be "not a straight-line-conserving model".
Prop 18 (p.8)	computed	Reproduced on 2×2 and the paper example. **Your email says this "proves the paper's ss:protrude claim"; the PDF (p.8, Gap 1) correctly says computed.** Fix the email/registry wording.

3. The main objection: $\sigma_W \circ C$ is a symmetry

Let C swap labels $0 \leftrightarrow 1$. My `tiles.py` (output in `tiles.log.txt`) finds, for all six reflections g: $g \circ C$ maps the KTW set onto itself. For the three reflections whose axis is an edge direction, which includes the wall reflection σ_W, additionally $(\sigma_W \circ C)$(hex) = hex *as a labelled tile*. (This is just your Cor. 5 plus hex$^R = C$(hex), which is visible in your p.2 table but never stated.) So the bulk set KTW $\cup$ {hex} has

an orientation-reversing symmetry fixing W; it acts on the wall data by $\sigma(\lambda) \mapsto$ its complement, i.e. by Grassmannian duality. That is exactly the situation of a "soliton non-preserving" / twisted-Yangian boundary, where the reflection equation is written with R conjugated by charge conjugation. Hence:

- Prop. 12 and Prop. 13 rule out only an *untwisted* fold. Remark 14's "there is no reflection symmetry at the wall for a reflection equation to express" (p.6) is false once C is allowed.
- More basically, the reflection equation $K_2 R_{21} K_1 R_{12} = R_{21} K_1 R_{12} K_2$ needs no geometric P-symmetry of R at all; lines reflect, the lattice does not have to. Even without C, Props 12/13 are arguments against a folding-trick construction, not against a K-matrix.

Requested fix: restate P2's verdict as "REFUTED for the untwisted fold; open for a C-twisted boundary; no R-matrix presentation exists yet to test either", which matches your own P1 finding that there is no R to test. Honestly the defensible verdict on P2 is *ill-posed*, same as P1.

4. Smaller points

- p.4 l. (proof of Cor 5): "the m even reflections" / "the m odd ones": write "reflections with m even"; as written m reads as a count.
- p.2: "seventeen tiles" but the p.2 table has 12 and the five extra are only named on p.8; Prop. 16's "ten of seventeen" lists eight names (the two relabelled 1-triangles are implicit). List them.
- p.3: the census-domain paragraph is good; please add the nonzero counts for the 208/208 rows as you did on p.8 (my 187-row has 44 nonzero, 54-row 16 nonzero; 168/187 for the wrong tile shows the zeros carry most of the agreement).
- Rem. 17 / §12: your own honeycomb remark is the right next step and undercuts Prop. 16's headline; promote it.

5. Lean Q559 boundary note

Verdict noted: `main_without_card_hypothesis_is_false` derives `False` from the converse stated without $r \le |M|$, witness $r = 1$, $M = \emptyset$, $\theta = 0$. Sensible from the statement: ψ is a sum over r-subsets of M (equivalently head-injections $[r] \hookrightarrow M$), which is empty for $|M| < r$, so the sum is 0 while ρ-shift injectivity is automatic for a single slot ($r = 1$); the claimed non-vanishing is then false. The $r = 0$ qualifier is correct but trivial ($|M| < 0$ never holds); I would not headline it. The point that 424/424 could not see the hypothesis because the generator drew $n \in [r, r+2]$ is the real lesson and is well taken. Grade of the node: proved (Lean-verified summand), as you state; parent unchanged.